

A kernel-writing agent under an 8k-token budget

Demo notes, Team 27 (seat 130) • Project 02, Stage A

2026-10-10 • <https://github.com/LiYou22/kernel-agent> • PR #20

1. The problem and our claim

Qwen3-8B (vLLM on a Trainium seat pod, 8192 tokens per call including the answer) must write tiled NumPy kernels for a 10-level ladder (relu ... conv1d) under chip-like rules: tiles of at most 128×512 , explicit loops, no fancy indexing or broadcasting, no calling the op itself, partial tiles handled. The dilemma: rules in the prompt make the small model audit itself into silence; no rules and it writes confident violations (our first L1 kernel: 40/40 correct, but `np.asarray` on line 5, so zero).

Claim: the rules belong in the verifier, not the prompt; every verdict is translated into one concrete instruction. The model writes freely from a one-sentence task; the harness finds what is wrong; the agent sends back the previous code, *one* instruction and one line of evidence; a kernel counts only if it also passes held-out cases the model never saw.

2. The verifier

- **Two rule checks:** a static AST scan (covers branches the tests never reach) and a runtime guard (numpy swapped for a monitored proxy that records each violation by line).
- **Numerical correctness** against a proven float32 error bound (§5), on hostile dev cases (huge magnitudes, large means, partial tiles) and a separate holdout set.
- **Diagnosis, not just a verdict:** cross-case patterns (“wrong exactly when there is more than one column tile”), and level hypotheses checked numerically against the reference before they are said (a wrong explanation is worse than none).
- **Cost ledger:** FLOPs, instructions, HBM bytes read/written for every run (§6).
- **Final gate:** the organizers’ `kernelbench.py`; its failures are translated too.

3. Key result: only the feedback differs

Same verifier, model, stopping rule and holdout; 5 runs per level; **ours** translates the verdict into one instruction, **naive** sends the verifier’s report back verbatim (version e22ccab872). Reward per run = best attempt: 0.1 parses + 0.2 obeys rules + 0.2 runs + $0.5 \times$ fraction of cases correct (the upstream NKI agent’s formula).

Level	solved (ours / naive)	per-run reward, ours	per-run reward, naive	mean reward
L1 relu	5/5 / 3/5	1.00 1.00 1.00 1.00 1.00	0.80 0.80 1.00 1.00 1.00	1.00 / 0.92
L2 row sum	4/5 / 0/5	1.00 1.00 1.00 0.80 1.00	0.80 0.80 0.80 0.80 0.80	0.96 / 0.80
L3 row max	5/5 / 1/5	1.00 1.00 1.00 1.00 1.00	1.00 0.80 0.81 0.80 0.80	1.00 / 0.84
L4 RMSNorm	1/5 / 0/5	0.94 0.61 0.61 1.00 0.94	0.74 0.61 0.61 0.61 0.74	0.82 / 0.66
Total	15/20 / 4/20			0.95 / 0.81

Naive’s many 0.80s are numerically right but rule-breaking (e.g. `np.sum`): given the raw report, the model repeats the same mistake. Tokens per verified kernel: **4.4k** vs **17.5k**, about 4× cheaper per success.

One failure and its recovery (L2, live log). #1 uses `np.sum`: 28/28 correct but illegal → “accumulate column by column, `acc += x[i0:i1, j]`”. #2 column loop, but indexes a 188-column partial tile with the global column: `IndexError: index 512 is out of bounds for axis 1 with size 188` (19/28 cases crash) → one instruction about the partial tile. #3 dev 28/28, holdout 21/21.

One new diagnosis (L3). The live bug `result[row] = acc` sat inside the column-tile loop, so each tile overwrote the last. The verifier now recomputes the reference per tile, sees the wrong outputs equal the last tile’s result, and says “keep one running value per row across all tiles; write it after the loop”. L3 went from stuck to 5/5.

4. Iteration backed by data

- **v3, pre-registered:** before any result we wrote “ $\geq 2/9$ on L4–L6 counts as better”. v3 scored 3/9 vs v2’s 1/25 (L6 0 → 2/3), so v3 was adopted. Caveat: six changes shipped together.
- **v4, aligned with the judges:** re-checking every kernel with the organizers’ `kernelbench.py` showed L4/L8 signature mismatches and float32 products failing its per-element 10^{-4} . v4 uses their signatures and makes their checker the final gate.
- **Final agent** (f0c39f1e0f, 74 self-tests) = v4 + a teammate’s v3-rewrite: a draft with ≥ 5 violations gets one instruction to rewrite in the op’s legal loop shape; “stuck” only when nothing improves; a red-team fix (column slices longer than 128 rows were not caught).

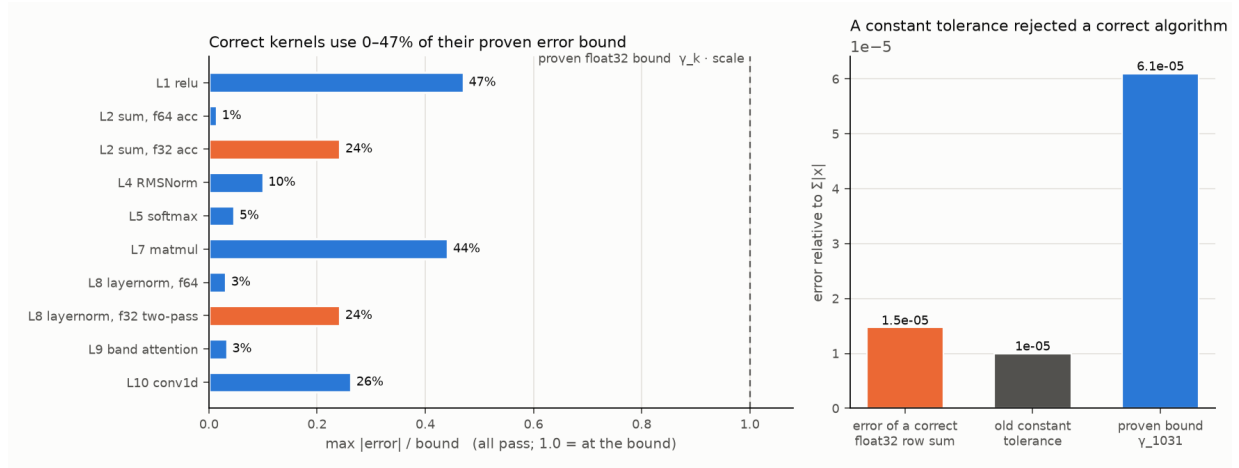
Final kernel per level, each written by the agent and passing *both* our checker (rules, dev, holdout) and the organizers’ `kernelbench.py` (FINAL.md, `kernels/final/`):

	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
both checkers	✓	✓	✓	–	✓*	✓	–	✓	✓*	✓
roofline time	1.00	1.00	1.00		2.00	1.00		2.00	61.2	12.2

* from a teammate’s run still in progress (version c5e98d5a75); L8 from the final agent, 1/3 runs. L4 and L7: no kernel passes both yet.

5. Numerical analysis: bound the error, don’t guess a tolerance

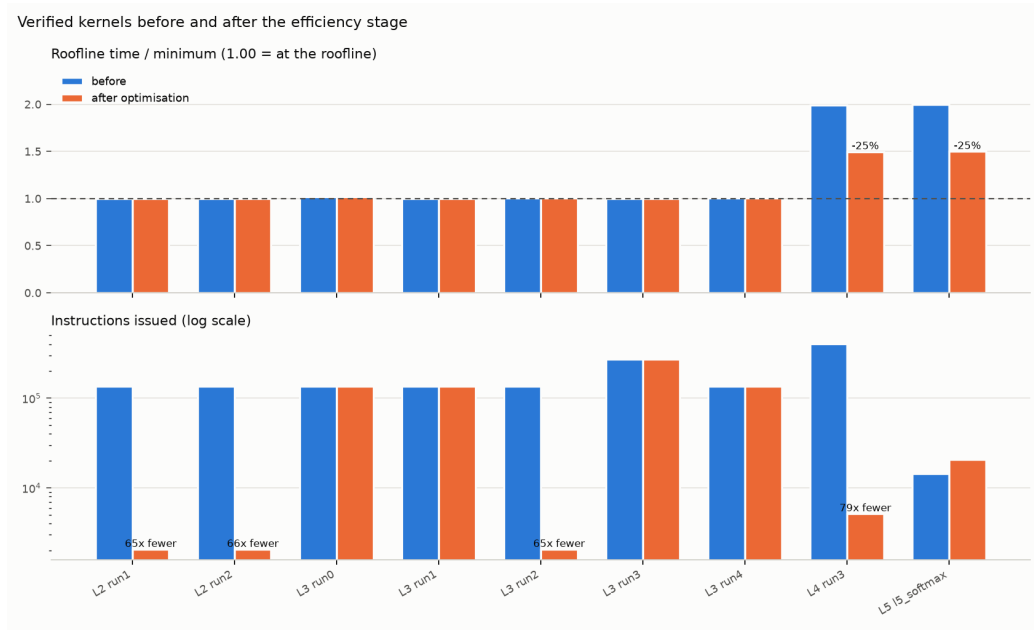
An output passes iff $|y - y_{\text{ref}}| \leq \gamma_k \cdot \text{scale}$ with $\gamma_k = ku/(1 - ku)$, $u = 2^{-24}$ (Higham), k the longest chain of roundings (row length for a sum, K for matmul, the window for attention/conv), scale the op’s natural magnitude (e.g. $\sum |x|$ for a row sum). Any algorithm correct in float32 passes, whatever its order; bugs still fail by orders of magnitude.



Our first constant tolerance (10^{-5}) rejected a *correct* float32 row sum (error 1.5×10^{-5} ; proven bound 6.1×10^{-5}). All hand kernels use 0–47% of their bound.

6. Efficiency: count FLOPs, bytes and instructions

The runtime guard keeps books for every run: FLOPs, instructions (numpy calls, including scalar ops), HBM bytes read and written, each against the op’s minimum. Cost is the roofline time $t = \max(\text{HBM bytes}, \text{FLOPs}/222)$ as a multiple of the minimum (222 FLOP/byte is the ridge). These ops run at 0.25–0.44 FLOP/byte, so they are memory-bound: the win is fewer passes and instructions, not fewer FLOPs.



8 of 15 verified model kernels were per-element loops ($\sim 135k$ instructions where column vectors need 2,068), all “correct”. The optimiser gives one rule-safe instruction with a snippet and accepts a rewrite only if dev *and* holdout still pass and it is cheaper: **65× / 79× fewer instructions**, L4 roofline $1.99\times \rightarrow 1.50\times$; no kernel was ever made wrong.

7. Honest limits

- L4 and up are still hard: two passes + no broadcasting + numeric traps; fixing one breaks another. L4 and L7 have no kernel passing both checkers.
- v3's six changes were shipped together, so the gain cannot be attributed to one of them; samples are small (3–5 runs) and the server is near-deterministic.
- The 8k budget never bound in Stage A (largest prompt 706 tokens); it would in Stage B (NKI).

8. Demo instructions (about 3 minutes, no model needed)

```
cd kernel-agent && uv sync
export KERNELBENCH_DIR=/path/to/02-kernel-agent # organizers' kernelbench.py (final gate)

# (a) the verifier names the cause, not just "wrong" -> Sec. 3, L3
uv run python -m kagent.harness 3 kernels/broken/l3_overwrite_per_tile.py --fixes
# (b) a rule violation becomes ONE instruction -> Sec. 3, L2
uv run python -m kagent.harness 2 kernels/broken/l2_np_sum.py --fixes
# (c) error bound and cost ledger on a correct kernel -> Sec. 5, 6
uv run python -m kagent.harness 4 kernels/hand/l4_rmsnorm.py
# (d) the full repair loop, replaying recorded model replies -> Sec. 1, 3
uv run python -m kagent.agent --levels 1 2 --scripted tests/scripts/demo.json --out /tmp/demo.jsonl
# (e) the per-run rewards of the ablation table -> Sec. 3
uv run python scripts/reward.py results/seat-130/ours-L1234-seat-130.jsonl \
    results/seat-130/naive-L1234-seat-130.jsonl
```

What to point at: (a) 0 rule violation(s), 15/24 cases correct and Pattern: ... equals the result over the last column tile only; (b) 28/28 cases correct yet np.sum [sum-like], then the FIX line 12 instruction; (c) worst uses 10% of it and the Cost line; (d) L2 FAIL rule:sum-like → PASS dev; holdout 21/21 → verified (confidence 0.95), with in=/out= token counts per attempt. Reproduce everything: `uv run pytest -q` (74 passed); README.md has the commands for live runs.